ShopSmart Retail Analytics - Part 1: Data Understanding

1. How many datasets were provided?

Nine datasets were provided: Customers, Products, Orders, Stores, Returns, Employees, Suppliers, Inventory, and Order Items. An Excel Data Dictionary was also provided to explain the fields in the datasets.

2. What does each dataset represent?

Each dataset contains information about a different part of the business. Customers contains customer details, while Products provides information about product names, categories, brands, and prices. Orders records sales transactions, and Stores contains information about store locations. Returns tracks returned items and refunds. Employees contains staff information, while Suppliers provides supplier contact and location details. Inventory shows stock levels and reorder levels, and Order Items contains details of the products in each order, including quantities, discounts, and line totals.

3. What is the primary key of each table?

The primary keys are Customer_ID, Product_ID, Order_ID, Store_ID, Return_ID, Employee_ID, Supplier_ID, Inventory_ID, and Order_Item_ID, respectively. These fields were checked for blank and duplicate values, and no issues were found in the checks performed.

4. What are the possible foreign keys?

Some ID fields are used to connect the datasets. Orders contains Customer_ID, Product_ID, and Store_ID, which can link it to Customers, Products, and Stores. Returns contains Order_ID, linking it to Orders, while Employees contains Store_ID, linking employees to their stores. Inventory contains Product_ID and Store_ID, connecting stock records to Products and Stores. Order Items contains Order_ID and Product_ID, linking it to Orders and Products.

The available fields did not provide a confirmed direct link between Suppliers and Products. The listed ID relationships were checked and reported as matching, although the relationship between Orders and Order Items needs further investigation to confirm how the records relate.

5. What is the relationship between the tables?

The datasets are connected mainly through their shared ID fields. Customers, Products, and Stores are linked to Orders, while Returns and Order Items are linked to Orders. Employees and Inventory are connected to Stores, and Inventory is also linked to Products.

Suppliers could not be directly connected to Products using the available columns. Also, Orders and Order Items each contain 50,000 records, so further checks are needed to understand their exact relationship. Understanding these connections is important because it helps ensure that the data can be combined and analysed correctly.